

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CE) Examination

November 2013

CE303.01 Computer Graphics

Date: 27.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Answer the following questions.

[06]

1. What is Computer Graphics? List out the applications of it.
2. Explain DVST.
3. Define i. Vertical Retrace ii. Aspect Ratio.

(b) Explain Refresh Cathode-Ray Tube with all components in detail.

[05]

Q - 2 (a) Explain Boundary Fill algorithm and differentiate 4-Connected and 8-Connected methods.

[04]

(b) Explain two basic techniques for producing color displays with CRT.

[04]

(c) Explain Scan Line Polygon Fill algorithm. What is the disadvantage of the algorithm? Write down the method to solve the disadvantage.

[04]

OR

Q - 2 (a) Calculate the points nearest to line path using DDA Line drawing algorithm. Line endpoints are: $(-4, -2)$ and $(4, 7)$.

[03]

(b) Answer the following questions.

[05]

1. Differentiate Raster Scan Displays and Random Scan Displays.

[02]

2. What is coherence property? How is it useful in scan-line polygon fill algorithm?

[02]

3. Differentiate Emissive and Non emissive display devices.

[01]

(c) Derive Bresenham's Line Drawing algorithm for $|m| < 1$ and write down the equations for $|m| > 1$.

[04]

Q - 3 (a) Explain Raster Scan System in detail.

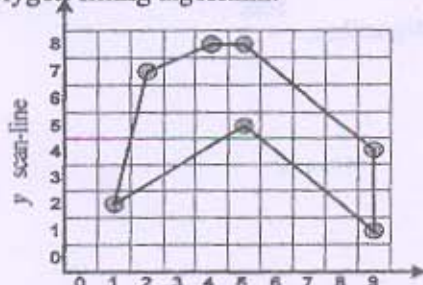
[04]

(b) Derive Midpoint circle drawing algorithm for the octant where slope of the curve varies from 0 to -1 .

[04]

(c) For a given polygon, prepare the Edge Table (ET) and Active Edge Table (AET) used in scan-fill polygon filling algorithm.

[04]



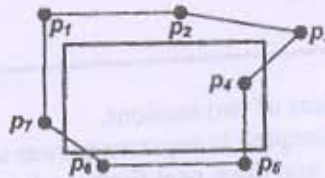
OR

(c) Derive Midpoint Ellipse algorithm from $(r_x, 0)$ to slope $= -1$.

[04]

SECTION - II

- Q - 4 (a) Prepare the list of vertices used to clip the polygon using Sutherland-Hodgeman polygon clipping algorithm against the rectangular clipping window. [04]



- (b) Do as Directed. [07]

1. Prove that the multiplication of 2D transformation matrices for each of the following sequence of operations is commutative. [03]
 - i. Two successive rotations.
 - ii. Two successive translations.
2. A line end points are A (0, 6) and B (6, 0). Reflect an object having vertices P(5, 3), Q (5, 5), R (3, 5), S (3, 3) with respect to given line. [04]

- Q - 5 (a) Briefly explain 3D translation and scaling. [05]

- (b) Do as Directed. [07]

1. Two end-points of a line: A (10, 45) and B (90, 25). Rectangle Clip window is (30, 20) and (75, 50). Clip the line using Liang-Barsky line clipping algorithm. [04]
2. What are the disadvantages of Cohen-Sutherland line clipping algorithm and compare it with Nicholl-Lee-Nicholl Line Clipping algorithm. [03]

OR

- Q - 5 (a) What is 2D shear transformation? Covert the unit square to shifted parallelogram using x-direction shear transformation operation where parameter $sh_x = \frac{1}{2}$ and $y_{ref} = -1$ and unit square dimensions are (0, 0), (1, 0), (0, 1) and (1, 1). [04]

- (b) Answer the following questions. [08]

1. Derive the composite matrix to perform general pivot-point (x_r, y_r) rotation in clockwise direction. [03]
2. Explain the term region codes. Write the steps of Cohen-Sutherland line clipping algorithm. [05]

- Q - 6 (a) Explain Nicholl-Lee-Nicholl line clipping algorithm. [06]

- (b) Explain RGB Model in detail. Which are the applications of it? [06]

OR

- Q - 6 (a) Explain Back-Face Detection algorithm. [06]

- (b) Explain Depth-Buffer method. [06]
